SIMATS ENGINEERING

ASSESSMENT TOOL 2-Pseudocode Writing Task

COURSE NAME: R PROGRAMMING **COURSE CODE: ITA04**

COURSE OUTCOME COVERED

CO1: Apply R programming fundamentals including data structures, vectors, matrices, arrays, and class types to perform mathematical operations and manage data effectively. Identify and implement appropriate control structures to solve a particular programming problem
(Bloom Level -BL3- BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 Questions **Total Marks: 100**

Code of Conduct:

I T.Saipraneeth *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

T. Saipraneeth
Signature of the student:

Pseudocode

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R Programming

1. Campus Canteen Billing

The Campus Canteen stores item prices in a numeric vector price and the quantity ordered in a vector qty for one customer. Write pseudocode to compute the bill for each item, the total bill, and apply a 10% discount if the total exceeds ₹500. Display final payable amount

```
START
input price, qty
bill = price*qty
total = SUM(bill)
IF total > 500
    discount = total * 10%
ELSE
    discount = 0
END IF

final = total - discount
Display final
  STOP
```

2. Student Mark Analyser

A teacher stores the marks of 40 students in a vector Write pseudo co to find (a) class average (b) the highest a lowest mark and (c) the count of studen who scored below 40 (fail). print "class performance: Good" if more than 80% have passed, otherwise "Needs improvement".

```
START
input marks
avg = MEAN(marks)
high = MAX(marks)
low = MIN(marks)
fail = COUNT(marks<40)
IF (40-fail)/40*100>80
  Display "Good"
ELSE
   Display "Needs improvement"
END IF
STOP
```

3. Hostel Room Allocation Matrix

A hostel has 4 floors and 5 rooms per floor. A 4x5 matrix stores if a room is occupied and 0 if vacant. Write

1. Aim: To write an R-pseudo code to display (a) the number of vacant rooms on each floor (b) the total vacant rooms in the hostel and (c) the floor with the maximum vacancies

```
START
input 4x5 matrix
Vacant = COUNT(0) in each row
total = SUM(Vacant)
floor = MAX(Vacant)
Display Vacant, total, floor
STOP
```

4. Monthly Electricity Bill calculator

An electricity board charges ₹3/unit for the first 100 units, ₹5/unit for next 100 units and ₹8/unit beyond 200 units. Given the units consumed by a house hold, write pseudo code to compute the total bill. Add a fixed charge ₹50 and print the bill summary.

```
START
input units
IF units <= 100
    bill = units*3
ELSE IF units <=200
    bill = 300 + (units-100)*5
ELSE
    bill = 800 + (units-200)*8
```

```
    END IF
bill = bill+50
Display bill
STOP
```

5. Employee Data Frame processing

An HR department maintains a data frame with columns EmpID, Name, Department, Salary Write pseudo code to (a) display employees from the "IT" department earning above ₹50,000 (b) Compute the avg Salary per department and (c) give 10% raise to employee's earning below ₹30,000 and add a new column New Salary.

```
START
input data frame
Display Department="IT" AND
                    Salary>50000

avg = AVERAGE Salary BY Department

FOR each employee
IF Salary < 30000
   New Salary = Salary * 1.10
ELSE
    New Salary = Salary
ENDIF

Display data frame
STOP
```

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode *Structure*	10%	Well-structured, *clear, consistent format, easy to follow*	Mostly clear *with minor issues*	Difficult to *follow in parts*	Poorly *structured and unclear*
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

